

functional Requirement and non-functional Requirement

functional Requirement :

Functional Requirement defines the specific features and operation a system must perform to meet business and user need. They describe what the system should do and how it interacts.

⇒ Focus on behaviour, and functional of system

⇒ Represent the feature that can be directly observed.

Ex: user authentication, data processing, search functionality

Non-functional Requirement :

non-functional requirement defines how a system should operate, focusing on performance, reliability, and user experience rather than specific features. They ensure the system is efficient, secure and maintainable.

Ex:

⇒ Performance

⇒ Security

⇒ Reliability

⇒ Scalability

⇒ maintainability

⇒ portability.

Boundary Value Analysis:

Boundary Value Analysis (BVA) is a black box technique that focuses on the testing the edges of input range to detect the commonly seen at the boundary values, which is used to identify error at the boundaries of input domain rather than in the middle of the range.

The core principle is that defects are more likely to occur at the edges of input range due to off-by-one error, incorrect condition (or) misconfiguration limit.

ex: Age validation:

Accept ages from 18 to 56.

valid $\rightarrow$ 18, 19, 29, 55, 56.

invalid $\rightarrow$ 0, 1, 9, 90, 92

Reason

here we have the limit 18 and 56 are limit 90, 92 are not in a range of given.